How Do We Engage with Other Disciplines? A Framework to Study Meaningful Interdisciplinary Discourse in Scholarly Publications

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Abstract

With the rising popularity of interdisciplinary work and increasing institutional incentives in this direction, there is a growing need to understand how resulting publications incorporate ideas from multiple disciplines. Existing computational approaches, such as affiliation diversity, keywords, and citation patterns, do not account for how individual citations are used to advance the citing work. Although, in line with addressing this gap, prior studies have proposed taxonomies to classify citation purpose, these frameworks are not well-suited to interdisciplinary research and do not provide quantitative measures of citation engagement quality. To address these limitations, we propose a framework for the evaluation of citation engagement in interdisciplinary Natural Language Processing (NLP) publications. Our approach introduces a citation purpose taxonomy tailored to interdisciplinary work, supported by an annotation study. We demonstrate the utility of this framework through a thorough analysis of publications at the intersection of NLP and Computational Social Science.

1 Introduction

Interdisciplinary research involves the integration of concepts, perspectives, theories, methods, and data from two or more bodies of specialized knowledge to construct creative solutions (National Academy of Sciences et al., 2005). Meta-analyses of how interdisciplinarity manifests in the scientific literature are therefore essential for understanding the convergence of, and interplay between, different fields. This definition highlights knowledge integration as a central component of interdisciplinarity, suggesting that it should be assessed not only in terms of its extent but also the quality with which ideas are meaningfully combined.

However, much of the existing meta-analytic work has focused on broad, scalable indicators (e.g., diversity and entropy of citations) (Porter and Rafols, 2009; Van Noorden, 2015; Cassi et al., 2017; Qun and Menghui, 2023) that emphasize the quantity and diversity of knowledge sources, but offer limited insight into how deeply or substantively those sources are integrated. For example, an article may cite a broad set of disciplines in its Introduction or Related Work sections without substantively engaging with them, such as by explicitly building on their ideas or using them to inform methodological choices. To truly understand the depth of engagement across fields, it is necessary to look beyond traditional measures and assess the role of interdisciplinary citations within the context of the citing work.

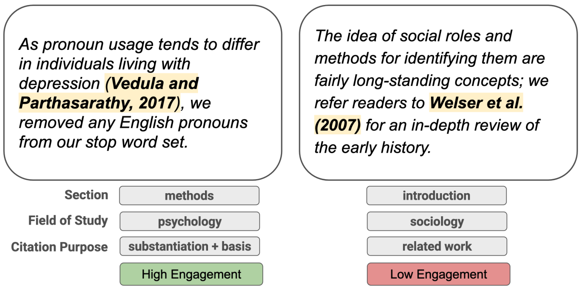

Figure 1: Framework for evaluating a publication's level of engagement with citing work.

Motivated by these challenges, we propose an approach to derive meaningful insights into the nature of interdisciplinary citations building on the framework of Leto et al. (2024). Starting from the hypothesis that the section in which a citation appears signals different levels of engagement, their framework extracts in-text citations from interdisciplinary papers along with the section in which they occur and an indicator of whether the cited work is out-of-discipline relative to the citing paper. We extend this framework by introducing a novel taxonomy to represent the purpose of an author in including a given citation. Although numerous citation-purpose classification schemes have been proposed (Jurgens et al., 2018; Abu-Jbara et al., 2013; Lauscher et al., 2022; Cohan et al., 2019; Teufel et al., 2006; Pride and Knoth, 2020), existing taxonomies are insufficient for analyzing interdisciplinary citations, either because they lack relevant categories, include categories that are ill-suited to cross-disciplinary contexts, or fail to incorporate adequate contextual information.

Our taxonomy addresses these limitations by situating each citation within its local argumentative context and characterizing citation purpose through categories designed to reflect the different forms of interdisciplinary engagement that arise when authors draw on work beyond their primary field. The taxonomy was developed from the ground up through an inductive annotation study, allowing categories to emerge from close analysis of real citation practices rather than being imposed a priori.

We ground our taxonomy development, annotation study, and analysis in the context of Natural Language Processing (NLP) research, positioning these contributions as a first step toward a broader framework for studying interdisciplinary engagement. We focus specifically on work at the intersection of NLP and Computational Social Science (CSS). Although research in this area has grown substantially over the past decade (Grimmer and Stewart, 2013), the quality of NLP+CSS publications has been questioned, with critiques arguing that such work often falls short of social science standards of rigor (Baden et al., 2022) and makes limited substantive contributions to those fields (McCarthy and Dore, 2023). In this work, we use our framework to further examine these concerns and to evaluate the extent to which citation engagement can be reliably studied at scale, including through the use of automated methods.

We make the following contributions[1]: (1) We develop a novel annotation framework designed to capture citation purpose in interdisciplinary NLP work. (2) We present an annotation study of the citation purpose taxonomy and a resulting annotated dataset of 394 annotated in-text citations taken from NLP+CSS work. (3) We demonstrate the usefulness of our annotation framework through an analysis of our manually-annotated dataset. (4) We evaluate the capacity of state-of-the-art language models to automatically annotate citations according to our taxonomy and identify key limitations that must be addressed to enable a fully automated pipeline for studying interdisciplinary engagement.

[1]Data can be found here.

2 Related Work

Numerous studies have introduced frameworks for classifying citation purpose or intent, but these approaches are not designed to capture how authors engage with references from outside their primary discipline. For example, Jurgens et al. (2018) and Pride and Knoth (2020) introduce datasets that are heavily dominated by a *Background* category designed to denote when a reference provides relevant information for the domain of the article. By treating all background citations as equivalent, this definition fails to distinguish between references that provide foundational knowledge necessary for interpreting the work and those that merely provide general or loosely related context. Similarly, their *Extension* category is narrowly defined to apply only when authors directly extend a method or dataset; a scenario that is relatively rare in interdisciplinary contexts, where engagement often takes other forms. In addition, their framework relies on a fixed citation context of approximately 300 surrounding characters, which may be insufficient to capture the broader argumentative role needed to accurately determine citation purpose. Lauscher et al. (2022) and Abu-Jbara et al. (2013) address this limitation by considering a window of sentences surrounding each citation — an approach that we adopt in this work.

Teufel et al. (2006) propose a highly fine-grained scheme for classifying citation function, with multiple sub-categories that distinguish forms of comparison and contrast. While this level of detail can be valuable, the resulting categories primarily capture evaluative stance and intra-discipline relations such as direct comparison and contrast. Additionally, their dataset labels over 60% of citations as *Neutral*, limiting its usefulness to distinguish meaningful forms of engagement. In contrast, our scheme places only 33% of citations into similar general categories, allowing for finer differentiation among types of interdisciplinary use.

At the opposite side of the spectrum, Cohan et al. (2019) introduce a taxonomy with only three high-level categories, which are too coarse-grained to adequately capture variation in the depth of citation engagement required for our analysis.

3 Data

In this section, we discuss our methods for collecting and processing our dataset of interdisciplinary NLP+CSS papers.

3.1 Interdisciplinary Article Collection

We construct a dataset of CSS papers from the ACL Anthology[2] published between 2014-2025. To do so, we build on the data collection methodology of Leto et al. (2024), who fine-tune an automatic track classifier for identifying NLP+CSS publications. They compile approximately 1,800 abstracts labeled by conference track using program information from 2021–2023. Papers categorized under "Computational Social Science and Cultural Analytics" serve as positive examples, while negative examples are drawn from all other tracks. We augment the positive set with papers from the ACL Special Interest Group on Language Technologies for the Socio-Economic Sciences and Humanities (SIGHUM) workshop and the Natural Language Processing and Computational Social Science (NLP+CSS). The resulting dataset comprises 497 positive and 2,692 negative examples, for a total of 3,189 labeled abstracts. The distribution of labels over time is shown in Figure 6. We use this labeled dataset to train a binary classifier to identify CSS papers. Specifically, we fine-tune RoBERTa-base (Gururangan et al., 2020) on abstract text. Details are provided in Appendix A.1.

We download all long publications in the ACL anthology from 2015-2025. Then, we use the classifier to identify 1,549 of the 17,258 unlabeled articles as NLP+CSS publications. Combined with the initial positively-labeled articles, this results in a dataset of 2,046 NLP+CSS articles.

3.2 Citation Mapping

To evaluate the engagement of a publication with the work it cites, we must first identify all in-text citations. To enhance the analysis, we are also interested in keeping track of the section they appear in, their field-of-study, and whether they are out-of-discipline.

After the CSS papers are identified, their content is processed using Grobid[3], then converted into dictionary representation using SciPDF Parser[4]. The resulting dictionary contains a breakdown of the textual content, including the reference section. Section names are mapped to canonical sections *Introduction*, *Related Work*, *Method*, *Experiments*, *Conclusion*, *References*, and *Appendix*. The in-text citations are then extracted from each section and assigned to their entry in the *References* section, connecting them to the title of the cited publication, the publication venue and other publication meta-data. Finally, reference entries are mapped to a "field of study" by querying the Semantic Scholar API (Kinney et al., 2023) or using a string-matching approach. All citations with a field of study outside of "Computer Science" and "Linguistics" are determined to be out-of-discipline.

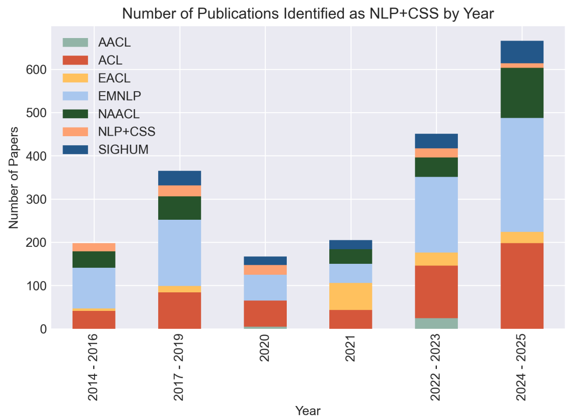

Figure 2: Dataset of NLP+CSS articles across years and venues

Our resulting dataset of interdisciplinary NLP+CSS work contains 2,052 articles (see Figure 2). From these, we extracted 37447 in-text citations with 11,158 interdisciplinary. We identified 26,289 of the in-text citations as out-of-discipline, for an average of out-of-discipline citations per article.

4 Engagement Framework

In this section, we discuss the novel taxonomy developed to classify citation purpose, and the corresponding annotation process for obtaining a manually-labeled subset.

4.1 Citation Purpose Taxonomy

Measuring the level of interdisciplinary engagement in a publication requires identifying the role that out-of-discipline references play in shaping the paper's conceptual, methodological, and empirical claims. For instance, a citation that contrasts a proposed approach with prior methods in the literature does not necessarily reflect deep engagement. In contrast, citations that reference methods or claims that are foundational to the analyses or arguments advanced in the published work indicate a substantially higher degree of engagement.

In line with this goal, we propose a citation-

[2] https://aclanthology.org/
[3] https://github.com/kermitt2/grobid
[4] https://github.com/titipata/scipdf_parser

Category	Explanation
SUBSTANTIATION +BASIS	The citation both substantiates a claim made in the citing work and serves as the basis for an idea or method that is subsequently built upon.
BASIS	The method, idea, or tool in the citation serves as the basis for the citing work. This must be explicitly stated or clearly evident from the abstract that the cited idea is central and explored further.
SUBSTANTIATION	The citation is used to verify or substantiate theoretical, empirical, or methodological claims made in the citing work. When classified as Substantiation, the supported claim should be identifiable.
USE	The method or tool in the citation is used directly in the citing work, with no or very trivial modifications.
DEFINITION	The citation is used to define or explain a topic or phrase (but not to validate it).
ANALYSIS	The method or idea in the citation is analyzed through comparisons or critiques that are directly linked to the citing work. The cited work may be compared to or criticized to justify an approach taken in the citing work, but is not explicitly built upon.
RELATED WORK	The citation refers to work that is related to the citing work or provides it as an example of a method, idea, or phenomenon discussed without further description. It does not fit any of the other categories.

Table 1: Citation purpose categories used for annotation.

purpose classification scheme adapted from Abu-Jbara et al. (2013), who define six categories: *Criticizing*, *Comparison*, *Use*, *Substantiating*, *Basis*, and *Neutral/Other*. Through an iterative, inductive annotation process conducted by two of the authors of the paper, we refine and extend this framework to better capture the roles that citations play in interdisciplinary contexts and to support assessments of engagement depth. The resulting taxonomy is presented in Table 1. Label determinations are made considering the paragraph surrounding each citation and the claims made in the paper's abstract, which together provide sufficient argumentative context for labeling decisions.

The categories *Basis* and *Substantiation + Basis* point to a deeper engagement with the literature. The latter, in particular, is frequently found in interdisciplinary work (see table 2). Often, a claim is validated by an out-of-discipline citation, then used to inform the proposed method, as exemplified in Figure 1. Another key feature of our taxonomy is the inclusion of the *Analysis* category, which combines the frequently-used categories *Comparison* and *Criticism* (Abu-Jbara et al., 2013). In most cases, these categories refer to the same general principle: analyzing the idea in the citation. Whether the ideas are being compared to or criticized does not signal different depths of engagement. The introduction of the *Definition* category helps identify citations that establish core concepts used in the work. Although citing a reference to support a definition may appear surface-level, such citations signal the conceptual importance of the defined term within the work and therefore provide meaningful evidence about the depth of interdisciplinary engagement.

Category	In Disc.	Out Disc.	Total
Substantiation + Basis	8	16	**24**
Basis	5	14	**19**
Substantiation	26	49	**75**
Use	46	18	**64**
Definition	5	6	**11**
Analysis	29	18	**47**
Related Work	53	76	**129**

Table 2: Number of annotated citations per citation purpose category with full or partial agreement.

Finally, we specifically design the *Related Work* category to include citations which support claims about the state of the literature (e.g. "There exist studies on this subject", "Multiple works have adopted this approach"). This distinction separates such citations from those that support empirical or findings-based claims, which are more appropriately categorized as *Substantiation*. In interdisciplinary work, the latter typically reflects a deeper and more consequential form of engagement than the former.

4.2 Annotation

To create our dataset, we performed an annotation study with three of the paper authors and two computer science graduate students. The graduate students received training from the paper authors and were provided with the Codebook included in

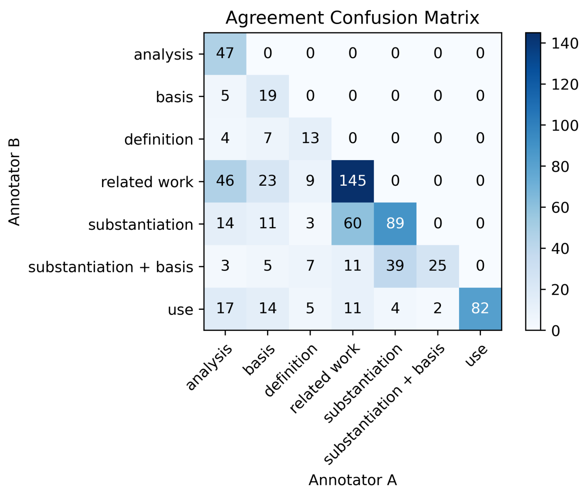

Figure 3: Types of agreements and disagreements among annotators.

Appendix A.3.

We selected about 170 articles balanced across publication year and venue. We selected 394 in-text citations from this set. 231 of the 394 examples were doubly-annotated, and 163 were triply-annotated. We had complete agreement on 64.21% of the citations and partial agreement on 93.91%. We obtained a Krippendorf's α of 0.6. The resulting agreed-upon dataset of 369 citations is summarized in Table 2. Agreements and disagreements between all annotators are presented in Figure 3.

4.3 Citation Engagement

We posit that the depth of engagement reflected by a citation is captured by three components: its purpose, the section of the paper in which it appears, and the semantic relatedness between the citation's local context and the claims articulated in the citing work's abstract. We operationalize this relatedness by computing the normalized cosine similarity between SPECTER embeddings (Cohan et al., 2020), which are trained in scientific documents and designed to capture semantic relationships between documents, for the citation context and the abstract. Tables 3 and 4 contain the citation sections and purposes, with a qualitative estimate of the depth of engagement that they capture.

5 NLP + CSS Engagement Analysis

In this section, we analyze our manually annotated dataset of 369 agreed-upon examples to demonstrate the utility of our proposed annotation and engagement framework.

Section	Engagement
Introduction	High
Method	High
Experiments	Medium
Appendix	Medium
Conclusion	Low
Related Work	Low

Table 3: Citation Section and Engagement

Citation Purpose	Engagement
Substantiation + Basis	High
Basis	High
Substantiation	Medium
Use	Medium
Definition	Medium
Analysis	Low
Related Work	Low

Table 4: Citation Purpose and Engagement

5.1 Distributional Patterns of Engagement

The distribution of citation purposes in our dataset is given in Table 2, and the distribution of sections in Table 5. We see that the majority of citations fall under *Related Work*, for both purpose and section, indicating only surface-level engagement in most cases. *Analysis* and *Related Work* together, indicating low engagement, make up 47% of the analyzed citations. Conversely, purposes indicating high engagement make up only 11% of the citations. We discuss further patterns in Appendix A.2.

An analysis of the relation between sections and purposes is presented in Figure 5, showing residuals resulting from a Chi-Square test. We obtain a p-value < 0.001, demonstrating that the association between paper section and citation purpose is highly statistically significant. Similarly, Figure 4 shows the distribution of relatedness scores between citation contexts and abstracts for each citation purpose. The limited variation in SPECTER similarity across citation purposes suggests that engagement cannot be reliably inferred from semantic similarity alone and must be considered in conjunction with citation purpose.

5.2 Qualitative Analysis of Engagement

We conduct a qualitative analysis of the annotated dataset to illustrate how different levels of engagement manifest in practice. Table 6 presents a representative subset of examples discussed in this

Section	In Disc.	Out Disc.	Total
Introduction	34	59	**93**
Related Work	40	46	**86**
Method	37	32	**69**
Experiments	24	21	**45**
Conclusion	13	19	**32**
Appendix	17	17	**34**

Table 5: Number of annotated citations per paper section with full or partial agreement.

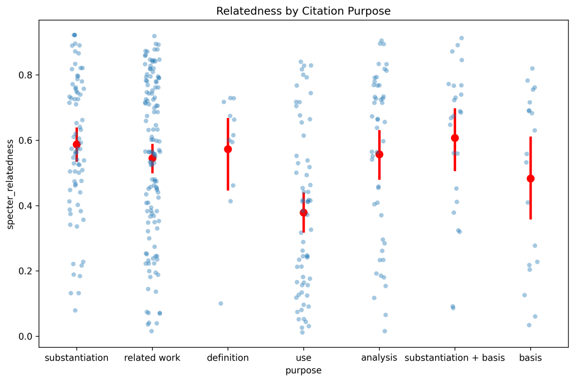

Figure 4: Correlation between citation purpose and context relatedness.

section, including excerpts from the citation context and the corresponding article abstracts. For each level of engagement, we highlight examples annotated with citation purposes aligned with that level, as defined in Table 4.

The citations that have high engagement with the citing work provide a conceptual basis for the article's work, or directly support the claims that the article makes in service of the method. For instance, the citation 'Ash et al., 2022,' supports the claim that intermediate text representations are useful for interpretative work, which serves as the basis for the article's methods, as indicated in the abstract by "...they rely on inferences that go beyond the observed language itself." Similarly, 'Hovy and Spruit, 2016' state that the opinions of dialect speakers may be mischaracterized in social media analyses, which motivates the article's goal of studying dialectal language. 'Bernard and Demeon, 2019,' offers a slightly different type of engagement, albeit just as deep — it provides a basis for studying and automating the studying of Byzantine Greek.

In contrast, citations that have low engagement usually only refer to work that is related to the topic at hand, but not significant to the contributions and outcomes of the citing work. This can be seen

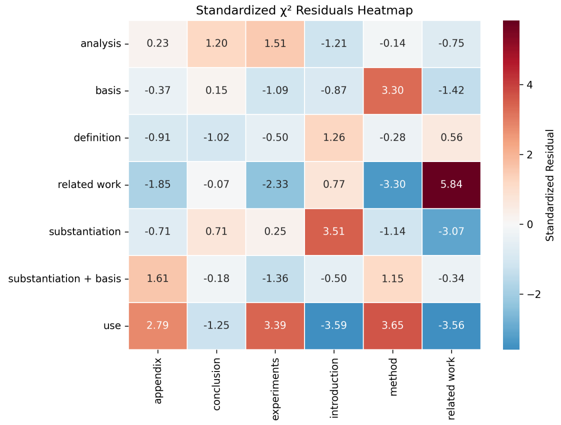

Figure 5: Correlation between citation section and purpose.

in the citations 'Yazdavar et al., 2020' and 'Cohen and Ruths, 2013,' which both merely focus on prior work done in the areas of their articles — work related to mental health, and work related to political views in tweets, which both only have shallow relationships with the claims made in their respective abstracts about the fairness of depression classifiers and the relationship between tweet author traits and their influence. 'Graham et al., 2009' has slightly higher engagement, in serving as a comparative baseline, but this is nevertheless still surface-level as the citing work does not meaningfully advance, adapt or discuss this work in relation to its own.

Citations with medium engagement are more nuanced. These contain works that are directly used in the citing article, or contain claims or explanations of terms relevant to the article's work. For instance, 'Papineni et al., 2002' refers to a metric that is used by the citing work, which shows that the citation is not superfluous or unnecessary; however, they do not engage with it critically or academically beyond using it. Similarly, 'Plank and Hovy, 2015' supports the claim that personality traits can be measured through linguistic behavior on social media, which does underlie the article's work. However, this is a broad claim, not specific to emojis or social media bios, and not used in any further manner. 'Sombart (1959)' offers a definition of a term, 'occupation', that is central to the article's work, but does not offer much to inform the task proposed.

Together, these examples reveal systematic differences in citation use across engagement levels: low-engagement citations primarily establish topical relevance, while medium-engagement citations

Engagement	Context	Citation Purpose
High	**Abstract**: When people interpret text, they rely on inferences that go beyond the observed language itself. Inspired by this observation, we introduce... **Context**: ...intermediate text representations are useful for interpretive work in the computational social sciences and digital humanities, where they can be aggregated to help uncover high-level themes and narratives in text collections (Bamman and Smith, 2015; **Ash et al., 2022**). In a similar vein, we cluster inferential decompositions of utterances...	Substantiation + Basis
High	**Abstract**: Though dialectal language is increasingly abundant on social media, few resources exist for developing NLP tools to handle such language. We conduct a case study of dialectal language in online conversational text by investigating African-American English (AAE) on Twitter... **Context:**...dialectal language is playing an increasingly prominent role in online conversational text, for which traditional NLP tools may be insufficient. This impacts many applications: for example, dialect speakers' opinions may be mischaracterized under social media sentiment analysis or omitted altogether (**Hovy and Spruit, 2016**). Since this data...	Substantiation + Basis
High	**Abstract**: This paper presents a pilot study to automatic linguistic preprocessing of Ancient and Byzantine Greek,and morphological analysis more specifically.... **Context**: The presented research is to be situated in the overarching DBBE project 4, where Byzantine epigrams are studied as nodes between textual transmission and cultural and linguistic developments (**Bernard and Demoen, 2019**). This multidisciplinary project...	Basis
Medium	**Abstract**: ...In this work, we focus on the use of emoji in these bio statements. We explore the ways in which users include emoji in these self-descriptions, finding different patterns than those observed around emoji usage in tweets. **Context**: ...Therefore, social media provides a natural opportunity to study self-identity. Previous studies have shown that specific personality characteristics can be measured by analyzing linguistic behavior on social media using natural language processing techniques **(Plank and Hovy, 2015)**.	Substantiation
Medium	**Abstract**: Since algorithms require concrete instructions, either in the form of rules or annotated examples, these abstract concepts must be broken down as part of the operationalisation process. In our paper, we tackle the task of identifying occupational titles in the plenary protocols of the German Bundestag. **Context**: **Sombart (1959**), for example, divides the term into an objective meaning, which focuses on the social function, and a subjective meaning, which emanates from the individual. He defines the social function of occupations very broadly, so that, for example, husband also becomes a profession.	Definition
Medium	**Abstract**: This paper examines Gemini's performance in translating an 18th-century Ottoman Turkish manuscript, Prisoner of the Infidels... **Context**: We report Bilingual Evaluation Understudy or BLEU scores **(Papineni et al., 2002)**	Use
Low	**Abstract**: This work provides an explanatory view of how LLMs can apply moral reasoning to both criticize and defend sexist language. **Context**: ...this distinction aligns with the reported divide between progressive and traditional views on the social roles of women in society, explained by MFT **(Graham et al., 2009)**	Analysis
Low	**Abstract**: ...Here, we analyze the fairness of depression classifiers trained on Twitter data with respect to gender and racial demographic groups. **Context**: Fortunately, this problem has been tackled using a multitude of different techniques across multiple studies specific to mental health (**Yazdavar et al., 2020**)...	Related Work
Low	**Abstract**: It has been claimed that people are more likely to be influenced by those who are similar to them than those who are not. In this paper, we test this hypothesis by measuring the impact of author traits on the detection of influence. **Context**: Predicting political orientation or ideologies has focused on predicting political views as left-wing vs right-wing in Twitter (Conover et al., 2011;**Cohen and Ruths, 2013**) or debates...	Related Work

Table 6: Selected examples of high, medium, and low levels of engagement with cited work and the corresponding citation purpose classification.

	DT + ChatGPT 5.2	Synth. + RoBERTa	Support
Substantiation + Basis	0	0.29	24
Basis	0.57	0.19	19
Substantiation	0.74	0.39	75
Use	0.75	0.58	64
Definition	0.40	0.35	11
Analysis	0.67	0.27	47
Related Work	0.42	0.03	129
Macro F1	**0.47**	0.30	369

Table 7: Per-class F1 performance of various models on test set of citations annotated with citation purpose taxonomy.

provide tools, definitions, or supporting assumptions without deeper critical engagement.

6 Toward Automatic Citation Purpose Classification

In this section, we assess the ability of state-of-the-art language models to automatically classify citation purpose according to the taxonomy introduced in Section 4. Specifically, we outline the model setup, evaluation, and discuss key limitations.

6.1 Models

Decision Tree + ChatGPT 5.2. We develop a decision tree-style prompt for purpose classification with ChatGPT 5.2 (OpenAI, 2025). See Appendix A.4.1 for prompt and additional details.

Synthetic Train Set + RoBERTa. We prompted Llama3-8b to generate 200 in-context example citations for each citation purpose category according to our codebook. We used 2-shot prompting with temperature set to 0.8. This synthetic set was then used to fine-tune RoBERTa base (Gururangan et al., 2020). See Appendix A.4.2 for additional details.

6.2 Evaluation

A summary of the results for the evaluated models is shown in Table 7. Overall, we find that neither approach achieves performance sufficient to support large-scale, fully automated analysis of citation purpose in scientific publications.

The **Synthetic Train Set + RoBERTa** model performs poorly across nearly all classes, with the exception of Use. We attribute this limitation primarily to the lack of diversity in the synthetically generated training data, as many generated examples closely mirror the provided prompts, resulting in limited coverage of the variation observed in real citation contexts. This issue is especially evident in the low F1 score for the *Related Work* category, which functions as a broad catch-all and exhibits substantial heterogeneity. We hypothesize that training on a sufficiently large, manually annotated dataset would substantially improve performance by providing a more representative and varied signal for fine-tuning. Alternatively, more sophisticated synthetic generation methods that introduce diversity could also improve performance.

Although the **Decision Tree + ChatGPT 5.2** approach substantially outperforms the synthetic supervised model overall, it nonetheless fails to correctly identify instances of the *Substantiation + Basis* category, yielding an F1 score of 0. This suggests that the model struggles to reliably distinguish citations that play a foundational or evidentiary role in supporting a paper's core claims, which are precisely the forms of engagement that are most critical for assessing interdisciplinary depth.

Taken together, these results highlight the current limitations of off-the-shelf language models. Realizing the vision of a fully automated framework for evaluating interdisciplinary engagement will likely require larger annotated training data, improved modeling of citation context and document-level claims, and closer alignment between model architectures and the conceptual distinctions encoded in our taxonomy.

7 Conclusion and Future Work

Motivated by the need to evaluate the engagement of interdisciplinary publications, we present a novel annotation taxonomy designed to capture citation purpose. Developed in the context of interdisciplinary NLP papers, the taxonomy includes categories specifically designed to capture interdisciplinary citation purpose with the downstream task of evaluating citation engagement. We present an annotation study of NLP+CSS work and a resulting dataset of 369 annotated citations. We include an analysis of our manually-annotated dataset, which reveals that the purpose of a citation is often related to the section it occurs in, and also that the purpose can be predictive of the level of engagement. These demonstrate the usefulness of our taxonomy for revealing insights about the field of NLP+CSS. As we develop and analyze this framework in the context of NLP work, it serves as a first step for the development of a broader framework to study interdisciplinarity. We leave the validation of this framework in additional fields to future work.

Limitations

This work has a number of limitations: (1) An automated framework was used to construct the dataset, potentially resulting in parsing errors and inaccuracies when extracting elements from scientific publications. (2) The presented analysis spans a limited number of articles and citations. Further analysis is required to understand whether the takeaways generalize to a larger dataset.

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A Appendix

A.1 CSS Track Identification Details

Labeled Dataset To fine-tune their automatic track classifier with the goal of obtaining a set of CSS papers, (Leto et al., 2024) collect a set of about 1,800 example abstracts labeled with their track using the conference programs from 2021-2023. Papers falling under the "Computational Social Science and Cultural Analytics" category serve as positive examples. Negative examples are collected from each additional track appearing in the program. Papers from the NLP+CSS workshop are appended to the positive set.

For a set of labels more representative of our dataset, we expand this dataset across 2014-2025 using this process. We also appended examples from the ACL Special Interest Group on Language Technologies for the Socio-Economic Sciences and Humanities (SIGHUM) workshop proceedings to the positive set. Note that no conference programs were available for 2024 and 2025, so the positive examples from this time were taken from the NLP+CSS and SIGHUM workshops. The resulting labeled dataset contains 497 positive examples and 2692 negative examples for a total of 3189 labeled examples. These are shown in Figure 6.

Model Details To build our binary track classifier, we fine-tune RoBERTa base (Gururangan et al., 2020) using the labeled abstracts using using HuggingFace's Transformers Trainer (Wolf et al., 2020), employing a weighted cross-entropy loss to address class imbalance. The model is trained for 50 epochs with a learning rate of $2e-5$, using an

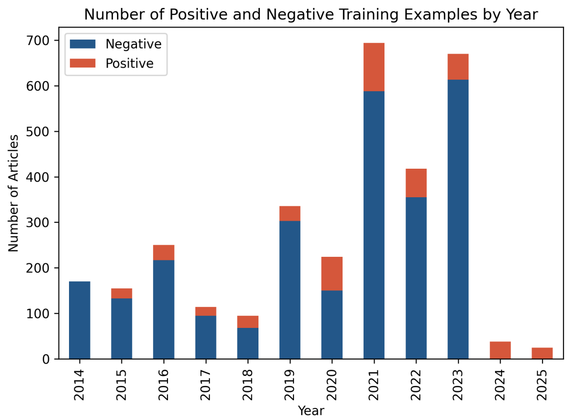

Figure 6: Labeled dataset for training track classifier

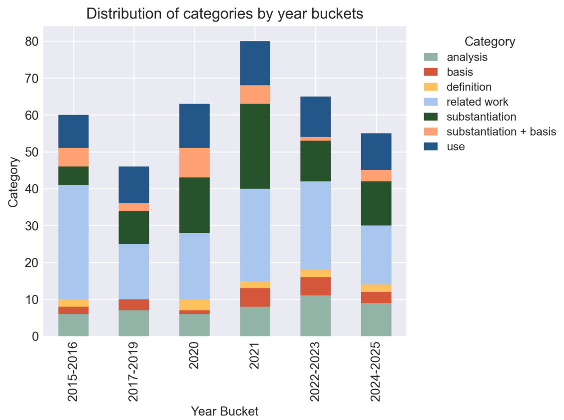

Figure 7: Distribution of citation purpose by article year.

80/10/10 train/validation/test split. The checkpoint achieving the highest macro F1 score on the validation set is selected, yielding a macro F1 of 0.870, precision of 0.863 and recall of 0.876 on the test set.

A.2 Additional Citation Purpose Annotation Details

Here, we present statistics on the distribution of citation purpose within the manually-annotated dataset across the citing work's year bucket (Figure 7) and venue (Figure 8).

A.3 Citation Purpose Codebook

The 7 categories below describe possible purposes for an out-of-discipline citation in an academic paper. Each category contains a description and guidelines for when the category is pertinent, a couple of examples, and an explanation of why that particular category is right for the example.

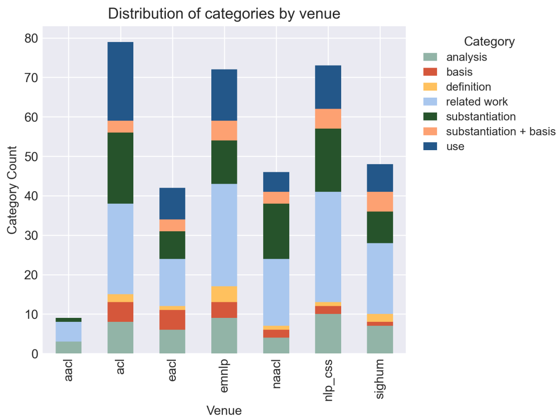

Figure 8: Distribution of citation purpose by article venue.

A.3.1 Use

The method or tool in the citation is used directly in the citing work, with no or very trivial modifications.

EXAMPLE

Abstract: In this work, we carry out a data archaeology to infer books that are known to ChatGPT and GPT-4 using a name cloze membership inference query. We find that OpenAI models have memorized a wide collection of copyrighted materials, and that the degree of memorization is tied to the frequency with which passages of those books appear on the web. The ability of these models to memorize an unknown set of books complicates assessments of measurement validity for cultural analytics by contaminating test data; we show that models perform much better on memorized books than on nonmemorized books for downstream tasks. We argue that this supports a case for open models whose training data is known.

Section: experiments

Citation Context: Our second experiment predicts the amount of time that has passed in the same 250-word passage sampled above, using the conceptual framework of **Underwood (2018)**, explored in the context of GPT-4 in Underwood (2023). Unlike the year prediction task above, which can potentially draw on encyclopedic knowledge that GPT-4 has memorized, this task requires reasoning directly about the information in the passage, as each passage has its own duration. Memorized information about a complete book could inform the prediction about all passages within it (e.g., through awareness of the general time scales present in a work, or the amount of dialogue within it), in addition to having exact knowledge of the passage itself. As above, our core question asks: do we see a difference in performance between books that GPT-4 has memorized and those that it has not? To assess this, we again identify the top and bottom decile of books by their GPT-4 name cloze accuracy. We manually annotate the amount of time passing in a 250-word passage from that book using the criteria outlined by **Underwood (2018)**, including the examples provided to GPT-4 in Underwood (2023), blinding annotators to the identity of the book and which experimental condition it belongs to. We then pass those same passages through GPT-4 using the prompt in figure 4, adapted from Underwood (2023). We calculate accuracy as the Spearman ρ between the predicted times and true times and calculate a 95% confidence interval over that measure using the bootstrap.

The above citation refers to a framework that is used directly in the citing work. There is no explicit reference to modifying this framework, or building upon it, and so it is inferred that the framework is used as is, and thus categorized as 'Use' and not 'Basis' (see category 6 below).

A.3.2 Definition

The citation is used to define a topic or phrase (but NOT validate it - see 'Substantiation').

EXAMPLE

Abstract: This paper analyses two hitherto unstudied sites sharing state-backed disinformation, Reliable Recent News (rrn.world) and WarOnFakes (waronfakes.com), which publish content in Arabic, Chinese, English, French, German, and Spanish. We describe our content acquisition methodology and perform cross-site unsupervised topic clustering on the resulting multilingual dataset. We also perform linguistic and temporal analysis of the web page translations and topics over time, and investigate articles with false publication dates. We make publicly available this new dataset of 14,053 articles, annotated with each language version, and additional metadata such as links and images. The main contribution of this paper for the NLP community is in the novel dataset which enables studies of disinformation networks, and the training of NLP tools for disinformation detection.

Section: introduction

Citation Context: Propaganda is defined as content that intentionally influences opinion to advance its creators' goals **(Bolsover and Howard, 2017)**. Numerous propaganda datasets have previously been created, with both document-level (Rashkin et al., 2017;Barrón-Cedeño et al., 2019) and span-level (Da San Martino et al., 2019b) technique annotations, using articles collected from multiple disinformation sites. At article-level, classifiers using combinations of multiple linguistic representations based on style and readability outperform content representation (Barrón-Cedeño et al., 2019), whereas content-based transformer models such as BERT have seen use at span-level (Da San Martino et al., 2019a). Detectors are often evaluated on single datasets, prompting concerns on generalisation (Martino et al., 2020).

The above citation offers a definition for "propaganda" that is used in the citing work.

A.3.3 Substantiation

The citation is used to verify or substantiate theoretical, empirical or methodological claims made in the citing work. Claims that describe the state of the literature (e.g. "There exist studies on this subject", "Multiple works have adopted this approach") are NOT valid claims for this category, and these citations might do better under 'Examples' or 'Related Work.' It does NOT matter if the claim is central to the premise of the citing work (such as claims made about contributions, or any claims in the abstract), or if the claim is local to the citation context. This category applies in any case where the citation is being used as proof of the claim. If classifying as 'Substantiation,' it is useful to identify the claim that is being supported.

EXAMPLE

Abstract: Prior work has revealed that positive words occur more frequently than negative words in human expressions, which is typically attributed to positivity bias, a tendency for people to report positive views of reality. But what about the language used in negative reviews? Consistent with prior work, we show that English negative reviews tend to contain more positive words than negative words, using a variety of datasets. We reconcile this observation with prior findings on the pragmatics of negation, and show that negations are commonly associated with positive words in negative reviews. Furthermore, in negative reviews, the majority of sentences with positive words express negative opinions based on sentiment classifiers, indicating some form of negation.

Section: introduction Citation Context: A battery of studies have validated the Pollyanna hypothesis that positive words occur more frequently than negative words in human expressions, using corpora ranging from Google Books to Twitter (Dodds et al., 2015;Garcia et al., 2012;Boucher and Osgood, 1969; **Kloumann et al., 2012**). The typical interpretation is connected with the positivity bias, which broadly denotes 1) a tendency for people to report positive views of reality, 2) a tendency to hold positive expectations, views, and memories, and 3) a tendency to favor positive information in reasoning (Carr, 2011; Augustine et al., 2011; Hoorens, 2014). However, it remains an open question whether the Pollyanna hypothesis holds in negative reviews, where the communicative goal is to express negative opinions.

The citation offers proof of the claim "positive words occur more frequently than negative words in human expressions." While the fact of this evidence is mentioned explicitly in the citation context above, this explicit mention is not necessary to apply this category to a citation.

A.3.4 Basis

The method, idea, or tool used in the citation is used as a basis for the method, idea, or tool in the citing work. It must be explicitly stated that this is being built upon, or it must be very evident from the abstract provided that this is an idea that is central to the premise and is thus being explored further.

EXAMPLE

Abstract: Online misogyny is a pernicious social problem that risks making online platforms toxic and unwelcoming to women. We present a new hierarchical taxonomy for online misogyny, as well as an expert labelled dataset to enable automatic classification of misogynistic content. The dataset consists of 6,567 labels for Reddit posts and comments. As previous research has found untrained crowdsourced annotators struggle with identifying misogyny, we hired and trained annotators and provided them with robust annotation guidelines. We report baseline classification performance on the binary classification task, achieving accuracy of

0.93 and F1 of 0.43. The codebook and datasets are made freely available for future researchers.

Section: method

Citation Context: This taxonomy draws on the typologies of abuse presented by Waseem et al. (2017) and **Vidgen et al. (2019)** as well as theoretical work in online misogyny research (Filipovic, 2007;Mantilla, 2013;Jane, 2016;Ging, 2017;Anzovino et al., 2018;Ging and Siapera, 2019;Farrell et al., 2019). It was developed by reviewing existing literature on online misogyny and then iterating over small samples of the dataset. This deductive-inductive process allowed us to ensure that conceptually distinct varieties of abuse are separated and that different types of misogyny can be unpicked. This is important given that they can have very different impacts on victims, different causes, and reflect different outlooks and interests on the part of the speaker.

The citation is classified as 'Basis' because the authors state that their taxonomy (central to their work, although this factor is not necessary) "draws on" the typology presented in the citation.

A.3.5 Substantiation + Basis

The citation is used as proof of a claim made in the citing work (see category 3 above), and this claim is either directly built upon in the citing work, or used as the basis for a specific idea (see category 4 above).

EXAMPLE

Abstract: Multiple studies have demonstrated that behavior on internet-based social media platforms can be indicative of an individual's mental health status. The widespread availability of such data has spurred interest in mental health research from a computational lens. While previous research has raised concerns about possible biases in models produced from this data, no study has quantified how these biases actually manifest themselves with respect to different demographic groups, such as gender and racial/ethnic groups. Here, we analyze the fairness of depression classifiers trained on Twitter data with respect to gender and racial demographic groups. We find that model performance systematically differs for underrepresented groups and that these discrepancies cannot be fully explained by trivial data representation issues. Our study concludes with recommendations on how to avoid these biases in future research.

Section: method

Citation Context: Tokenization. Raw text within in Tweets was tokenized using a modified version of the Twokenizer (O'Connor et al., 2010). English contractions were expanded, while specific retweet tokens, username mentions, URLs, and numeric values were replaced by generic tokens. As pronoun usage tends to differ in individuals living with depression **(Vedula and Parthasarathy, 2017)**, we removed any English pronouns from our stop word set (English Stop Words from nltk.org). Case was standardized across all tokens, with a single flag included if an entire post was made in uppercase letters.

The citation offers proof for the claim "pronoun usage tends to differ in individuals living with depression," and this claim is used to justify their removal of English pronouns in their approach.

A.3.6 Analysis

The method or idea in the citation is being analyzed in some way by presenting comparisons or critiques. These analyses must have a direct link with the citing work, as evidenced in the citation context or the abstract. For example, the citation's idea or method may be compared to one in the citing work, or may be criticized and its limitations used to justify an approach taken in the citing work. There must be no explicit mention of building on the citation's idea in the citing work.

EXAMPLE

Abstract: Natural language inference (NLI) is the task of determining whether a piece of text is entailed, contradicted by or unrelated to another piece of text. In this paper, we investigate how to tease systematic inferences (i.e., items for which people agree on the NLI label) apart from disagreement items (i.e., items which lead to different annotations), which most prior work has overlooked. To distinguish systematic inferences from disagreement items, we propose Artificial Annotators (AAs) to simulate the uncertainty in the annotation process by capturing the modes in annotations. Results on the CommitmentBank, a corpus of naturally occurring discourses in English, confirm that our approach performs statistically significantly better than all baselines. We further show that AAs learn linguistic patterns and context-dependent reasoning.

Section: method

Citation Context: If we view the AAs as a committee of three members, our architecture is reminiscent of the Query by Committee (QBC) **(Seung et al., 1992)**, an effective approach for active learning paradigm. The essence of QBC is to select unlabeled data for labeling on which disagreement among committee members (i.e., learners pre-trained on the same labeled data) occurs. The selected data will be labeled by an oracle (e.g., domain experts) and then used to further train the learners. Likewise, in our approach, each AA votes for an item independently. However, the purpose is to detect disagreements instead of using disagreements as a measure to select items for further annotations. Moreover, in our AAs, the three members are trained on three disjoint annotation partitions for each item (see Section 3.2).

The citation refers to an architecture that is similar to the one proposed in the citing work. It is clear from the citation context that the authors of the citing work are contrasting their own architecture with the one in the citation.

A.3.7 Related Work

he citation refers to work that is related to the citing work or provides it as an example of a method, idea, or phenomenon discussed without further description. It does not fit any of the other categories.

EXAMPLE

Abstract: Methods and applications are inextricably linked in science, and in particular in the domain of text-as-data. In this paper, we examine one such text-as-data application, an established economic index that measures economic policy uncertainty from keyword occurrences in news. This index, which is shown to correlate with firm investment, employment, and excess market returns, has had substantive impact in both the private sector and academia. Yet, as we revisit and extend the original authors' annotations and text measurements we find interesting text-as-data methodological research questions: (1) Are annotator disagreements a reflection of ambiguity in language? (2) Do alternative text measurements correlate with one another and with measures of external predictive validity? We find for this application (1) some annotator disagreements of economic policy uncertainty can be attributed to ambiguity in language, and (2) switching measurements from keyword-matching to supervised machine learning classifiers results in low correlation, a concerning implication for the validity of the index.

Section: experiments

Citation Context: KeyExp. Although Baker et al. use human auditors to find policy keywords that minimize the false positive and false negative rates, they do not expand or optimize for economy or uncertainty keywords. Thus, we expand these keyword lists via GloVe word embeddings 9 (Pennington et al., 2014), and find the five nearest neighbors via cosine distance. 10 This is a simple keyword expansion technique. In future work, one could look to the literature on lexicon induction to improve creating lexicons that represent the semantic concepts of interest (Taboada et al., 2011;Pryzant et al., 2018;**Hamilton et al., 2016**;Rao and Ravichandran, 2009). Alternatively, one could also create a probabilistic classifier over pre-selected lexicons to soften the predictions, or use other uncertainty lexicons or even automatic uncertainty cue detectors.

As inferred from the given context, the citation describes work related to creating lexicons that represent the semantic concepts of interest, which is identifiable as related to the citing work. This is classified as 'Related Work,' and not 'Examples,' because there are multiple ideas being discussed, and it is not clear which, if any, the citation is supposed to be an example of. Additionally, the ideas being referenced are potential future directions, implying a certain level of novelty; it is therefore unlikely that the citation contains a concrete example of the specified idea.

A.4 Purpose Classification Model Details

In this section we present further details including prompts used for the models described in Section 6.

A.4.1 Decision Tree + ChatGPT5.2

Prompts used for this method are shown in Figures 9 and 10.

A.4.2 Synthetic Train Set + RoBERTa

Prompts for generating the synthetic training set are shown in Figures 11 and 12. To build the classifier, we train on the full generated dataset, using 10% of

System Prompt

You are a classifier, identifying the purpose of a given citation. You will see a CSV with the citation text, citation context, abstract, and section. You must classify the citation text based on the context, of which it is a substring. You may use the abstract (of the citing work) and the section of occurrence only if needed. I will describe a decision tree to you. At each step, decide Yes/No, and proceed accordingly until you reach the leaves with the categories.

< decision tree >

Return a CSV as plain text with the citation ID, the category, and your reasoning. If classified as 'Substantation' or 'Substantiation + Basis,' write the claim in another column. Make sure to analysis the context specifically in relation to the given citation text; many categories may fit the context as a whole, or may fit other citations in the context, but you must make your decisions for the given citation text ONLY.

Figure 9: System prompt used to prompt ChatGPT 5.2 for citation purpose classification.

Decision Tree

Step 1: Does the context offer a critique of the citation's work? If yes, go to step 2.1. If NO, go to step 2.

Step 2: Is the citation's work being explicitly compared to the citing work or used as a baseline? If YES, go to step 2.1. If NO, go to step 3.

Step 2.1: Is there explicit mention of building on the citation's idea in the context? If YES, classify as SUBSTANTIATION + BASIS. If NO, classify as ANALYSIS.

Step 3: Is the citation's idea or method being used or built upon in the citing work? If YES, go to Step 3.1. If NO, go to step 4.

Step 3.1: Is there a theoretical or empirical claim associated with the idea or method that motivates its use? If YES, go to step 5.5. If NO, go to step 3.2.

Step 3.2: Is the citation referring to a conceptual idea and not a concrete method? If YES, classify as BASIS. If NO, go to step 3.3.

Step 3.3: Does the context explicitly state building upon, improving, or modifying the citation's method? If YES, classify as BASIS. If NO, classify as USE.

Step 4: Is the context offering a definition or explanation of a topic or phrase based on the citation? If YES, go to step 4.1. If NO, go to step 5.

Step 4.1: Does the definition or explanation include a justification, validation or proof of some kind? If YES, classify as SUBSTANTIATION. If NO, classify as DEFINITION.

Step 5: Is the citation providing evidence for a claim or assertion? The support does not have to be linguistically explicit. If YES, go to step 5.1. If NO, go to step 6.

Step 5.1: Does the claim describe the state of the literature (e.g. "There exist studies on this subject", "Multiple works have adopted this approach")? If YES, classify as RELATED WORK. If NO, go to step 5.2.

Step 5.2: Is the citation being used as proof of a theoretical, empirical, or methodological claim? If YES, go to step 5.3. If NO, classify as RELATED WORK.

Step 5.3: Is the claim being used as the basis for a specific idea in the citing work or being built upon in some way? If YES, classify as SUBSTANTIATION + BASIS. If NO, classify as SUBSTANTIATION.

Figure 10: Decision tree used to prompt ChatGPT 5.2 for citation purpose classification.

System Prompt

You are an annotator who is developing a synthetic dataset of examples for a taxonomy designed to identify the purpose for a publication to use a particular in-text citation.

Your answer should be in JSON using the following format: {'citation': '(author_last_name, publication_data)', 'context': 'paragraph-long context of the citation being used for the specified purpose'}. Note that 'citation' should be an exact substring of 'context'. Do not generate anything else.

Figure 11: System prompt used to generate synthetic training set for RoBERTa classifier.

User Prompt

Generate an example of a cited work being used for the purpose specified. Add a plausible, paragraph-long context (at least 5 sentences) around the example.

Citation Purpose: <purpose category>
<purpose category> Definition: <purpose category definition>

Figure 12: User prompt used to generate synthetic training set for RoBERTa classifier.

the training dataset for validation. We train for 10 epochs with a learning rate of $2e-5$. The model which obtained the maximum Macro F1 score on the validation set was loaded at the end of training and tested on the manually-annotated dataset.

	precision	**recall**	**f1-score**	**support**
substantiation + basis	0.207	0.353	0.261	17
basis	0.143	0.429	0.214	7
substantiation	0.526	0.488	0.506	41
definition	0.2	0.4	0.267	5
analysis	0.237	0.474	0.316	19
use	0.429	0.48	0.453	25
related work	0.75	0.056	0.103	54
accuracy	0.327	0.327	0.327	0
macro avg	0.356	0.383	0.303	168
weighted avg	0.493	0.327	0.303	168

Table 8: Detailed classification results of citation purpose RoBERTa classifier trained on LLM-generated synthetic dataset.